Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

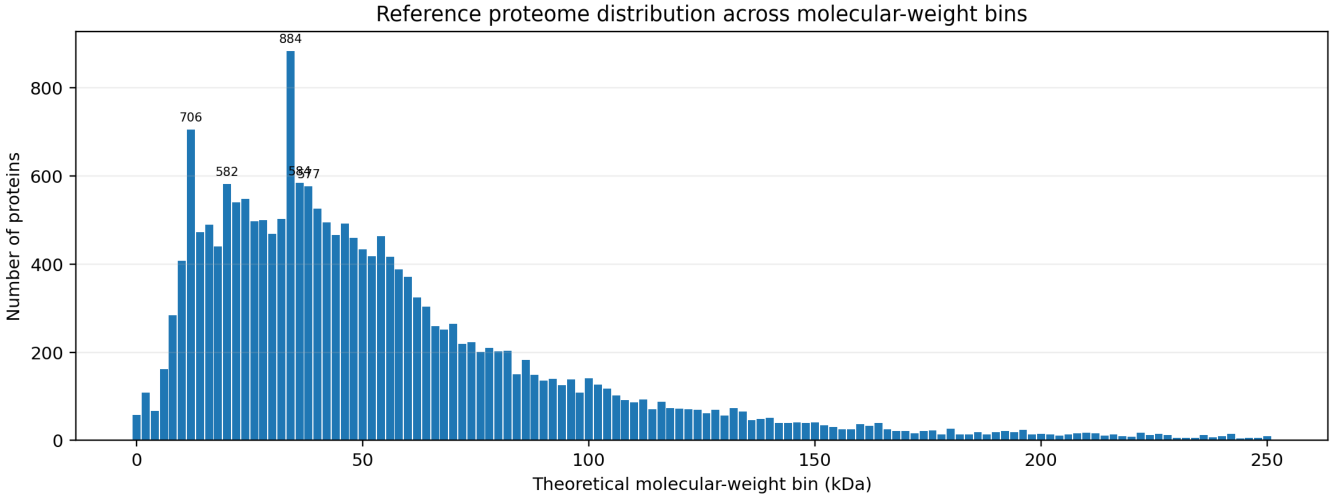
Metric	Value
Proteins parsed	20,652
Molecular-weight bins	273
Bin width	2 kDa
MW range	0.26 to 3816.03 kDa
Median MW	45.82 kDa
Mean MW	61.77 kDa
Proteins <= 50 kDa	11,317 (54.8%)
Proteins 50-100 kDa	6,283 (30.4%)
Proteins > 100 kDa	3,052 (14.8%)
High-priority bins	29
Elevated-priority bins	40
Routine-priority bins	204
Source FASTA	UP000005640_9606.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

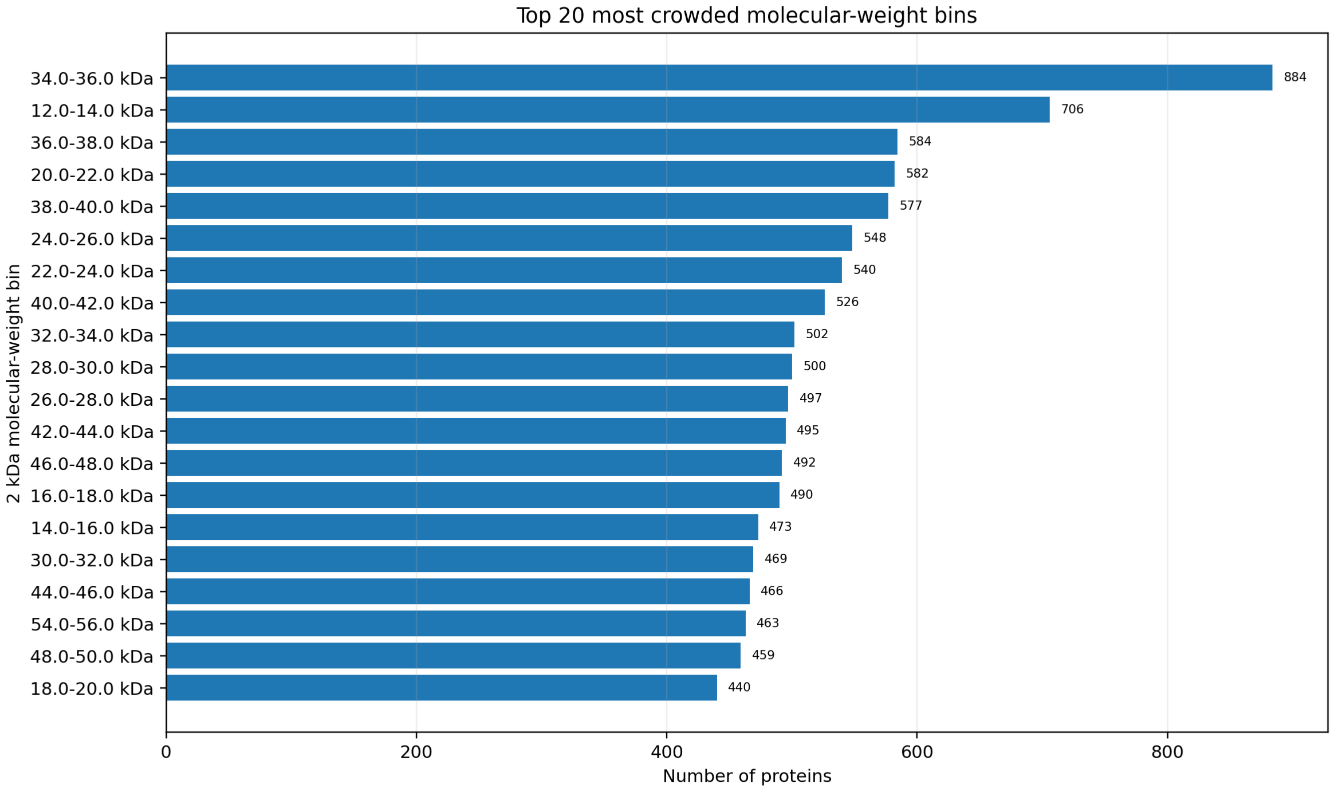
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



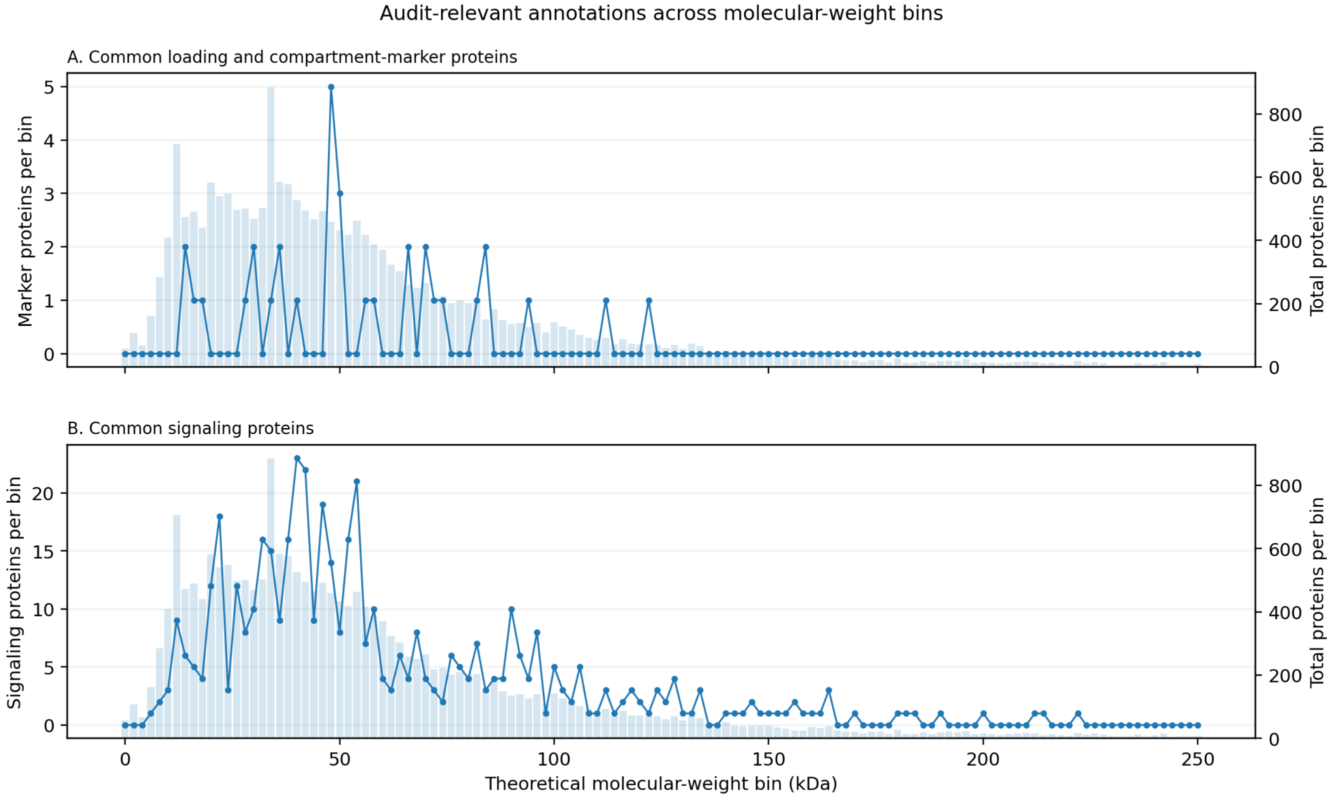
Most crowded molecular-weight neighborhoods.

The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



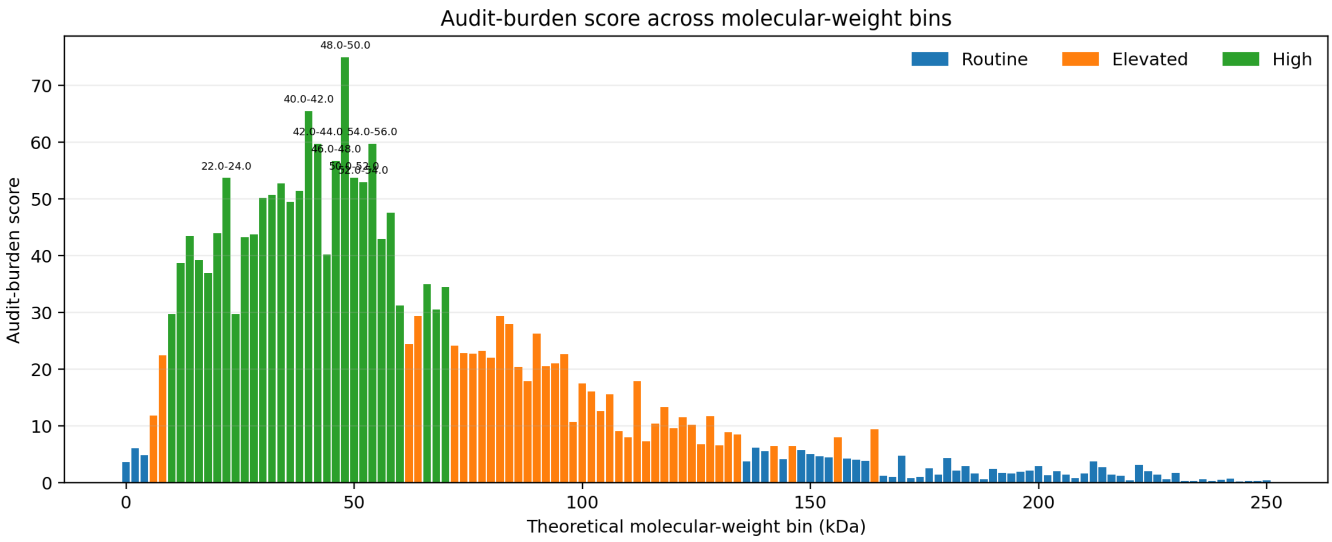
Distribution of common western blot markers and signaling proteins across molecular-weight bins.

The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
48.0-50.0 kDa	459	5	14	high	75.0
40.0-42.0 kDa	526	1	23	high	65.5
42.0-44.0 kDa	495	0	22	high	59.8
54.0-56.0 kDa	463	0	21	high	59.8
46.0-48.0 kDa	492	0	19	high	56.8
22.0-24.0 kDa	540	0	18	high	53.8
50.0-52.0 kDa	433	3	8	high	53.8
52.0-54.0 kDa	418	0	16	high	53.0
34.0-36.0 kDa	884	1	15	high	52.8
38.0-40.0 kDa	577	0	16	high	51.5
32.0-34.0 kDa	502	0	16	high	50.8
30.0-32.0 kDa	469	2	10	high	50.2
36.0-38.0 kDa	584	2	9	high	49.5
58.0-60.0 kDa	388	1	10	high	47.6
20.0-22.0 kDa	582	0	12	high	44.0
28.0-30.0 kDa	500	1	8	high	43.8
14.0-16.0 kDa	473	2	6	high	43.5
26.0-28.0 kDa	497	0	12	high	43.2
56.0-58.0 kDa	417	1	7	high	43.0
44.0-46.0 kDa	466	0	9	high	40.2
16.0-18.0 kDa	490	1	5	high	39.2
12.0-14.0 kDa	706	0	9	high	38.8
18.0-20.0 kDa	440	1	4	high	37.0
66.0-68.0 kDa	259	2	4	high	35.0
70.0-72.0 kDa	265	2	4	high	34.5

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
48.0-50.0 kDa	459	5	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	high	MRPS5, MOK, PSKH1, BZW1, BMP10, IFRD2, PANX1, ESRRB, KRT18, TFAP2A, RNF150, PLIN2, ZNF597, STAMBP, KPTN, SMAD3 [FOXO/SMAD transcription], NBPF7P, KRT17
40.0-42.0 kDa	526	1	23	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	CYTIP [Actin family], ANGPTL7, ELAC1, VN1R1, GLT8D2, CXADR, GNAT1, B4GALT4, PACC1, LHX6, POMK, GNAO1, PTGFR, L1RE1, HLA-E, MACROH2A2 [Histone family], NPIPA3 [Actin family], ERI1
42.0-44.0 kDa	495	0	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	SSTR4, ACTBL2 [Actin family], SPOUT1, ACTA2 [Actin family], TECRL, CRLF2, CCR9, POTEKP [Actin family], ACTC1 [Actin family], ZDHHC2, CCN1 [Cell-cycle CDK/cyclin], ZDHHC18, POLDIP2 [Actin family], DUSP5, ACTA1 [Actin family], FAM217B, HSD3B2, FETUB
54.0-56.0 kDa	463	0	21	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	MMP12, STK38L, MMP1, ZBTB45, GCGR, HCLS1, SLC16A4, PI4K2A, PFKFB4, METTL4, SIGLEC8, SLC38A1, SLC17A4, FUCA2, SLC2A1, CAMK2A, ADPGK, NARS2
46.0-48.0 kDa	492	0	19	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	ABHD3, OLFML3, TNFRSF19, EGLN1, B3GNT2, TNFRSF11B, HEPACAM, OAS1, PCID2, COX15 [Mitochondrial markers], TPBG, SEC14L3, ZC3H10, TDG, GPR137, AADACL4, POLDIP3 [Actin family], ACP4
22.0-24.0 kDa	540	0	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	RHOC, LIF, ZNF581, RAB1C, AP3S2, H1-6 [Histone family], PTTG1, ANKRD66, CCDC28B, C2orf80, PPIF [Mitochondrial markers], FGF4, TMEM223, IFNA14, PTTG3P, MSC, SNX22, STMN4
50.0-52.0 kDa	433	3	8	Actin family; Apoptosis caspase/BCL; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	TTC23, RASSF9, TINF2 [Actin family], SEPTIN14, FAM90A20, ONECUT3, FOLH1B, INTS15, PIGU, NKD2, PLA2G7, SIGLEC9, FAM90A2P, TNFRSF10A, TUBA8 [Tubulin family], ADSS2, MAP2K5 [MAPK cascade], SLC29A2
52.0-54.0 kDa	418	0	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	high	GLRA2, FAM81B, CHRDL1, ATP6AP1, TIGD3, PKNOX2, FEZF1, PPP2R2D, CPSF7, MYBPH, XKRX, CHST9, GTPBP3 [Mitochondrial markers], CELF1, AGT, GPAT4, PROC, MFSD12
34.0-36.0 kDa	884	1	15	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	FCN2, ZFP36, PILRA, PPCS, PGP, SLC25A1 [Mitochondrial markers], OR7A17, TP53TG5 [p53 pathway], OR1R1, OR9G9, RHBDL2, LRRC30, TMX2, OR5BS1, FCGR2B, CDK5R1 [Cell-cycle CDK/cyclin], SLC25A11 [Mitochondrial markers], PRSS29P
38.0-40.0 kDa	577	0	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	EFNB1, FST, OPCML, PAQR5, GIMAP2, PPP1R3G, CCDC42, MTARC2 [Mitochondrial markers], TAAR8, RRN3P2, ACMSD, TBPL2, MAGT1, DNAJB1, AQP7B, ST3GAL4, ST6GALNAC6, FAM228B
32.0-34.0 kDa	502	0	16	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	PGAM5 [Mitochondrial markers], PRSS42P, CUEDC2, YIF1A, KLK5, GJB1, HNRNPCL3, HNRNPCL4, GSX2, CCDC106, SCGN, ART5, ASRGL1, DGUOK [Mitochondrial markers], TAF1D, ZMAT3, SLC25A29 [Mitochondrial markers], RNASEH1
30.0-32.0 kDa	469	2	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; Tubulin family	high	BPGM, ASB13, HHEX, CALB1, PLSCR5, KRTAP10-9, COMT, EXOSC8, TLCD4, HAPSTR2, HLA-DRB5, NOTCH2NLR, TIGAR, NSA2, PIANP, PLP1, THAP8, CER1
36.0-38.0 kDa	584	2	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	LOC128966728, VGLL3, ZAR1L, ASB7, OR5A2, BNIP2 [Actin family], AKR1B10, KIR3DP1, HMOX2, TECR, MC3R, OR5I1, GIPC1, OR56B1, POU5F2, GAPDH, CASTOR2, OR5V1
58.0-60.0 kDa	388	1	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	LCK, RCOR2, OXSR1, IKZF3, GLCCI1, CCT6A, GALNS, ZBTB7B, POGLUT2, PAK2, SPNS2, ZRSR2, STK35, SLC29A4, SNTB1, CYP8B1, ZC3H12D, ZNF837
20.0-22.0 kDa	582	0	12	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	LINC02908, TMEM52B, COX4I2 [Mitochondrial markers], IL22, TTC9C, DCTD, FTL, VCX3A, SAT1, SMIM23, TIMM22 [Mitochondrial markers], MRPL17, , IL1RN, AK6, JPT2, COPRS, TMEM217B
28.0-30.0 kDa	500	1	8	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	CTRL, LRRC61, TSPAN3, FAM220A, ZFAND2B, CRYBB1, RPL8, HLA-DQA2, DPY19L2P1, KLK8, CMBL, SPR, NR0B2, HOXB9, , RTRAF, MS4A12, PDCL2
14.0-16.0 kDa	473	2	6	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; p53 pathway	high	CXCL9, H2AJ [Histone family], SRSP, GABARAPL1, PAGE5, SMR3A, TEX46, CLPSL1, BEX4, LIMD2, LSM10, SEM1, ADAMTSL4-AS1, H2AC11 [Histone family], TEX54, H2AC18 [Histone family], HCP5, H2AC6 [Histone family]
26.0-28.0 kDa	497	0	12	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Mitochondrial markers; Tubulin family; p53 pathway	high	TMED5, PRR23E, C1QA, TNFSF8, FBXL22, MOB4, SPAG7, TSPAN8, NMRK2, GINS4, CD79B, BCL2L1 [Apoptosis caspase/BCL], RND1, PBDC1, MXI1 [Actin family], RTP2, ATP6V1E2, SPEGNB
56.0-58.0 kDa	417	1	7	AKT pathway; Actin family; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	high	KCNS3, CYP21A2 [p53 pathway], IPPK, CDKL2 [Cell-cycle CDK/cyclin], SLC38A2, TENT2, ZNF525, LPCAT3, IRF5, PHF10, SAMD1, SLC9B1, ZNF639, ZBTB37, STEAP2, TPH2, MGAT4C, USP14
44.0-46.0 kDa	466	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers	high	SERPINE2, NIPAL4, NKX1-1, RIBC1, KCNJ9, AGPAT4, MEI4, ZNG1B, PAFAH2, ZNG1C, QTRT1, GNA13, B3GNT5, ERCC8, PDZK1P1, PIPOX, ZNG1E, ZNG1A
16.0-18.0 kDa	490	1	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	LEKR1, TUG1, C14orf119, JPT1, LINC00588, C1D, PXT1, HBD, LINC01565, RPS19, CSNK1G2-AS1, SMIM44, MRPL27, CST5, CCDC197, PLA2G2A, TMEM250, CNIH4
12.0-14.0 kDa	706	0	9	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	DNAL4, IGLV3-25, EMC6, C1orf210, CXorf51A, CXorf51B, PSENEN, PAGE2B, IGLV3-19, TRAV27, NAP1L6P, BLID, WFDC12, NDUFB2 [Mitochondrial markers], PVALB, BPY2, , WDR83OS
18.0-20.0 kDa	440	1	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	high	NCBP2, ZCCHC13, UBE2I, MTFP1 [Mitochondrial markers], KRTAP1-5, PPIA, CTAGE3P, BIK [Actin family], H2BC19P [Histone family], UMODL1-AS1, , KRTAP4-4, TIMM17A [Mitochondrial markers], IGF2-AS, TMEM238, ACP1, LCN6, IL17F
66.0-68.0 kDa	259	2	4	Actin family; Cell-cycle CDK/cyclin; ER markers; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	CKAP4, ZNF583, TNFRSF11A, IGFALS, KRT1, , NOP56, MBD4, FRRS1, FARSB, FSIP1 [Actin family], IGF2BP2, CHRM3, ENC1, SLC13A1, TRIP4, MPP3, ACSM3 [Mitochondrial markers]
70.0-72.0 kDa	265	2	4	Actin family; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	RUFY2, HSPA2 [Heat shock proteins], PABPC3, F2, CLGN, SLCO2A1, ZP1, HSPA1A [Heat shock proteins], HSPA1B [Heat shock proteins], NELFB, FNBP1L, NTN4, ZSWIM9, ENOX2, FBXL4, GLCE, MOCS1, F11
60.0-62.0 kDa	371	0	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	DCLRE1B, ZNF454, ZNF350, OTOA, CSPG5, OIT3, RBM46, KRT6C, UGT1A4, SLC22A7, KPNA6, GATA6, FMO3, ZNF215, RPS6KL1, KRT6A, SLC26A10P, KRT6B
68.0-70.0 kDa	251	0	8	Actin family; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	high	, PES1, SLC6A13, FPGT, PTPN11, PGBD2, PIAS3, PTPN9, SUSD5, PRODH [Mitochondrial markers], KLHL41, LRIT1, KARS1, GATAD2A, POF1B, LINGO2 [Actin family], MSLN, WDR46
24.0-26.0 kDa	548	0	3	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	FGF19, DPT, KLRF2, TNNI3, RNASE10, GCA, HOXB7, MISP3, SOST, HMGB2, PDGFA, TRIM61, GTF3C6, FRAT2, DNAAF6, TBC1D28, CREG1, CBLN2
10.0-12.0 kDa	408	0	3	Actin family; Apoptosis caspase/BCL; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers	high	FLVCR1-DT, DEFB109D, WT1-AS, DBI, LINC03043, BANF1, CENPW, MLLT11, DEFB115, DEFA5, UQCC3, CCL3, DPM3, CHCHD7, SCGB3A1, NPS, PYDC1, SRP9
82.0-84.0 kDa	204	1	7	Actin family; Apoptosis caspase/BCL; Heat shock proteins; MAPK cascade; Mitochondrial markers	elevated	SANBR, VWA2, ZBTB1, MZF1, ARHGAP28, FYB2, MX2, EEF2K, FIGN, LRRFIP2 [Actin family], DZANK1, SLC6A16, VPS52, DCLK1, INTS10, MTSS1, ORC3, EXT2

Common loading/compartment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
14.23	H2AC1	Q96QV6	H2A1A_HUMAN	Histone H2A type 1-A	14.0-16.0 kDa
15.40	H3C1	P68431	H31_HUMAN	Histone H3.1	14.0-16.0 kDa
16.30	TOMM20	Q15388	TOM20_HUMAN	Mitochondrial import receptor subunit TOM20 homolog	16.0-18.0 kDa
19.58	COX4I1	P13073	COX41_HUMAN	Cytochrome c oxidase subunit 4 isoform 1, mitochondrial	18.0-20.0 kDa
28.77	PCNA	P12004	PCNA_HUMAN	DNA sliding clamp PCNA	28.0-30.0 kDa
30.77	VDAC1	P21796	VDAC1_HUMAN	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
31.57	VDAC2	P45880	VDAC2_HUMAN	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
34.27	RPLP0	P05388	RLA0_HUMAN	Large ribosomal subunit protein uL10	34.0-36.0 kDa
36.05	GAPDH	P04406	G3P_HUMAN	Glyceraldehyde-3-phosphate dehydrogenase	36.0-38.0 kDa
37.70	TBP	P20226	TBP_HUMAN	TATA-box-binding protein	36.0-38.0 kDa
41.74	ACTB	P60709	ACTB_HUMAN	Actin, cytoplasmic 1	40.0-42.0 kDa
48.14	CALR	P27797	CALR_HUMAN	Calreticulin	48.0-50.0 kDa
49.67	TUBB	P07437	TBB5_HUMAN	Tubulin beta chain	48.0-50.0 kDa
49.90	TUBA1C	Q9BQE3	TBA1C_HUMAN	Tubulin alpha-1C chain	48.0-50.0 kDa
49.91	TUBB2A	Q13885	TBB2A_HUMAN	Tubulin beta-2A chain	48.0-50.0 kDa
49.95	TUBB2B	Q9BVA1	TBB2B_HUMAN	Tubulin beta-2B chain	48.0-50.0 kDa
50.14	TUBA1A	Q71U36	TBA1A_HUMAN	Tubulin alpha-1A chain	50.0-52.0 kDa
50.15	TUBA1B	P68363	TBA1B_HUMAN	Tubulin alpha-1B chain	50.0-52.0 kDa
50.43	TUBB3	Q13509	TBB3_HUMAN	Tubulin beta-3 chain	50.0-52.0 kDa
57.12	P4HB	P07237	PDIA1_HUMAN	Protein disulfide-isomerase	56.0-58.0 kDa
59.75	ATP5F1A	P25705	ATPA_HUMAN	ATP synthase F(1) complex subunit alpha, mitochondrial	58.0-60.0 kDa
66.41	LMNB1	P20700	LMNB1_HUMAN	Lamin-B1	66.0-68.0 kDa
67.57	CANX	P27824	CALX_HUMAN	Calnexin	66.0-68.0 kDa
70.05	HSPA1A	P0DMV8	HS71A_HUMAN	Heat shock 70 kDa protein 1A	70.0-72.0 kDa
70.90	HSPA8	P11142	HSP7C_HUMAN	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
72.69	SDHA	P31040	SDHA_HUMAN	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
74.14	LMNA	P02545	LMNA_HUMAN	Prelamin-A/C	74.0-76.0 kDa
83.26	HSP90AB1	P08238	HS90B_HUMAN	Heat shock protein HSP 90-beta	82.0-84.0 kDa
84.66	HSP90AA1	P07900	HS90A_HUMAN	Heat shock protein HSP 90-alpha	84.0-86.0 kDa
84.87	TFRC	P02786	TFR1_HUMAN	Transferrin receptor protein 1	84.0-86.0 kDa
95.34	EEF2	P13639	EF2_HUMAN	Elongation factor 2	94.0-96.0 kDa
112.90	ATP1A1	P05023	AT1A1_HUMAN	Sodium/potassium-transporting ATPase subunit alpha-1	112.0-114.0 kDa
123.80	VCL	P18206	VINC_HUMAN	Vinculin	122.0-124.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	58	0	0	Actin family	routine	TRDD1, TRBD1, IGHD1-1, IGHD5-12, IGHD5OR15-5B, IGHD6-19, IGHD6-13, IGHD3-10, , IGLJ4, IGHD3-9, IGKJ4, IGKJ3, IGKJ2, IGLJ5, IGHD3-3, IGKJ5, LOC128462377
2.0-4.0 kDa	108	0	0	Mitochondrial markers	routine	TRAJ22, TRAJ36, TRAJ50, TRAJ6, TRAJ33, TRAJ17, TRAJ41, TRAJ54, TRAJ9, , IGHJ2, TRAJ42, TRAJ5, TRAJ7, TRAJ57, TRAJ44, TRAJ26, TRAJ25
4.0-6.0 kDa	67	0	0	Actin family; Mitochondrial markers	routine	FAM86JP, ID2B, OST4, BAGE4, , MT-RNR2, LOC128071545, , IGLJ2, LOC128031832, , KRTAP20-4, , LOC128092246, BAGE5, IGLJ1, BAGE,
6.0-8.0 kDa	162	0	1	JAK-STAT; Mitochondrial markers; p53 pathway	elevated	PRCD, MT1E, PMAIP1, TOMM5 [Mitochondrial markers], MT1H, MT2A, MT1L, MT1X, MT1F, MT1HL1, SMIM30, ARHGAP5-AS1, PLN, MT1M, MT1B, MT1A, MT1G, HTN3
8.0-10.0 kDa	284	0	2	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	DEFB110, TOMM6 [Mitochondrial markers], GNG12, UQCC6 [Mitochondrial markers], GTF2H5, PIGY, DEFB124, TMEM167A, NDUFA1, CD24, CCDC179, DEFB123, C16orf74, RBAKDN, TEX53, COX8C [Mitochondrial markers], DEFB131B, SPRR2G
10.0-12.0 kDa	408	0	3	Actin family; Apoptosis caspase/BCL; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers	high	FLVCR1-DT, DEFB109D, WT1-AS, DBI, LINC03043, BANF1, CENPW, MLLT11, DEFB115, DEFA5, UQCC3, CCL3, DPM3, CHCHD7, SCGB3A1, NPS, PYDC1, SRP9
12.0-14.0 kDa	706	0	9	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	DNAL4, IGLV3-25, EMC6, C1orf210, CXorf51A, CXorf51B, PSENEN, PAGE2B, IGLV3-19, TRAV27, NAP1L6P, BLID, WFDC12, NDUFB2 [Mitochondrial markers], PVALB, BPY2, , WDR83OS
14.0-16.0 kDa	473	2	6	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; p53 pathway	high	CXCL9, H2AJ [Histone family], SRSP, GABARAPL1, PAGE5, SMR3A, TEX46, CLPSL1, BEX4, LIMD2, LSM10, SEM1, ADAMTSL4-AS1, H2AC11 [Histone family], TEX54, H2AC18 [Histone family], HCP5, H2AC6 [Histone family]
16.0-18.0 kDa	490	1	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	LEKR1, TUG1, C14orf119, JPT1, LINC00588, C1D, PXT1, HBD, LINC01565, RPS19, CSNK1G2-AS1, SMIM44, MRPL27, CST5, CCDC197, PLA2G2A, TMEM250, CNIH4
18.0-20.0 kDa	440	1	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	high	NCBP2, ZCCHC13, UBE2I, MTFP1 [Mitochondrial markers], KRTAP1-5, PPIA, CTAGE3P, BIK [Actin family], H2BC19P [Histone family], UMODL1-AS1, , KRTAP4-4, TIMM17A [Mitochondrial markers], IGF2-AS, TMEM238, ACP1, LCN6, IL17F
20.0-22.0 kDa	582	0	12	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	LINC02908, TMEM52B, COX4I2 [Mitochondrial markers], IL22, TTC9C, DCTD, FTL, VCX3A, SAT1, SMIM23, TIMM22 [Mitochondrial markers], MRPL17, , IL1RN, AK6, JPT2, COPRS, TMEM217B
22.0-24.0 kDa	540	0	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	RHOC, LIF, ZNF581, RAB1C, AP3S2, H1-6 [Histone family], PTTG1, ANKRD66, CCDC28B, C2orf80, PPIF [Mitochondrial markers], FGF4, TMEM223, IFNA14, PTTG3P, MSC, SNX22, STMN4
24.0-26.0 kDa	548	0	3	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	FGF19, DPT, KLRF2, TNNI3, RNASE10, GCA, HOXB7, MISP3, SOST, HMGB2, PDGFA, TRIM61, GTF3C6, FRAT2, DNAAF6, TBC1D28, CREG1, CBLN2
26.0-28.0 kDa	497	0	12	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Mitochondrial markers; Tubulin family; p53 pathway	high	TMED5, PRR23E, C1QA, TNFSF8, FBXL22, MOB4, SPAG7, TSPAN8, NMRK2, GINS4, CD79B, BCL2L1 [Apoptosis caspase/BCL], RND1, PBDC1, MXI1 [Actin family], RTP2, ATP6V1E2, SPEGNB
28.0-30.0 kDa	500	1	8	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	CTRL, LRRC61, TSPAN3, FAM220A, ZFAND2B, CRYBB1, RPL8, HLA-DQA2, DPY19L2P1, KLK8, CMBL, SPR, NR0B2, HOXB9, , RTRAF, MS4A12, PDCL2
30.0-32.0 kDa	469	2	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; Tubulin family	high	BPGM, ASB13, HHEX, CALB1, PLSCR5, KRTAP10-9, COMT, EXOSC8, TLCD4, HAPSTR2, HLA-DRB5, NOTCH2NLR, TIGAR, NSA2, PIANP, PLP1, THAP8, CER1
32.0-34.0 kDa	502	0	16	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	PGAM5 [Mitochondrial markers], PRSS42P, CUEDC2, YIF1A, KLK5, GJB1, HNRNPCL3, HNRNPCL4, GSX2, CCDC106, SCGN, ART5, ASRGL1, DGUOK [Mitochondrial markers], TAF1D, ZMAT3, SLC25A29 [Mitochondrial markers], RNASEH1
34.0-36.0 kDa	884	1	15	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	FCN2, ZFP36, PILRA, PPCS, PGP, SLC25A1 [Mitochondrial markers], OR7A17, TP53TG5 [p53 pathway], OR1R1, OR9G9, RHBDL2, LRRC30, TMX2, OR5BS1, FCGR2B, CDK5R1 [Cell-cycle CDK/cyclin], SLC25A11 [Mitochondrial markers], PRSS29P
36.0-38.0 kDa	584	2	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	LOC128966728, VGLL3, ZAR1L, ASB7, OR5A2, BNIP2 [Actin family], AKR1B10, KIR3DP1, HMOX2, TECR, MC3R, OR5I1, GIPC1, OR56B1, POU5F2, GAPDH, CASTOR2, OR5V1
38.0-40.0 kDa	577	0	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	EFNB1, FST, OPCML, PAQR5, GIMAP2, PPP1R3G, CCDC42, MTARC2 [Mitochondrial markers], TAAR8, RRN3P2, ACMSD, TBPL2, MAGT1, DNAJB1, AQP7B, ST3GAL4, ST6GALNAC6, FAM228B
40.0-42.0 kDa	526	1	23	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	CYTIP [Actin family], ANGPTL7, ELAC1, VN1R1, GLT8D2, CXADR, GNAT1, B4GALT4, PACC1, LHX6, POMK, GNAO1, PTGFR, L1RE1, HLA-E, MACROH2A2 [Histone family], NPIPA3 [Actin family], ERI1
42.0-44.0 kDa	495	0	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	SSTR4, ACTBL2 [Actin family], SPOUT1, ACTA2 [Actin family], TECRL, CRLF2, CCR9, POTEKP [Actin family], ACTC1 [Actin family], ZDHHC2, CCN1 [Cell-cycle CDK/cyclin], ZDHHC18, POLDIP2 [Actin family], DUSP5, ACTA1 [Actin family], FAM217B, HSD3B2, FETUB
44.0-46.0 kDa	466	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers	high	SERPINE2, NIPAL4, NKX1-1, RIBC1, KCNJ9, AGPAT4, MEI4, ZNG1B, PAFAH2, ZNG1C, QTRT1, GNA13, B3GNT5, ERCC8, PDZK1P1, PIPOX, ZNG1E, ZNG1A
46.0-48.0 kDa	492	0	19	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	ABHD3, OLFML3, TNFRSF19, EGLN1, B3GNT2, TNFRSF11B, HEPACAM, OAS1, PCID2, COX15 [Mitochondrial markers], TPBG, SEC14L3, ZC3H10, TDG, GPR137, AADACL4, POLDIP3 [Actin family], ACP4
48.0-50.0 kDa	459	5	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	high	MRPS5, MOK, PSKH1, BZW1, BMP10, IFRD2, PANX1, ESRRB, KRT18, TFAP2A, RNF150, PLIN2, ZNF597, STAMBP, KPTN, SMAD3 [FOXO/SMAD transcription], NBPF7P, KRT17
50.0-52.0 kDa	433	3	8	Actin family; Apoptosis caspase/BCL; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	TTC23, RASSF9, TINF2 [Actin family], SEPTIN14, FAM90A20, ONECUT3, FOLH1B, INTS15, PIGU, NKD2, PLA2G7, SIGLEC9, FAM90A2P, TNFRSF10A, TUBA8 [Tubulin family], ADSS2, MAP2K5 [MAPK cascade], SLC29A2
52.0-54.0 kDa	418	0	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	high	GLRA2, FAM81B, CHRDL1, ATP6AP1, TIGD3, PKNOX2, FEZF1, PPP2R2D, CPSF7, MYBPH, XKRX, CHST9, GTPBP3 [Mitochondrial markers], CELF1, AGT, GPAT4, PROC, MFSD12
54.0-56.0 kDa	463	0	21	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	MMP12, STK38L, MMP1, ZBTB45, GCGR, HCLS1, SLC16A4, PI4K2A, PFKFB4, METTL4, SIGLEC8, SLC38A1, SLC17A4, FUCA2, SLC2A1, CAMK2A, ADPGK, NARS2
56.0-58.0 kDa	417	1	7	AKT pathway; Actin family; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	high	KCNS3, CYP21A2 [p53 pathway], IPPK, CDKL2 [Cell-cycle CDK/cyclin], SLC38A2, TENT2, ZNF525, LPCAT3, IRF5, PHF10, SAMD1, SLC9B1, ZNF639, ZBTB37, STEAP2, TPH2, MGAT4C, USP14
58.0-60.0 kDa	388	1	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	LCK, RCOR2, OXSR1, IKZF3, GLCCI1, CCT6A, GALNS, ZBTB7B, POGLUT2, PAK2, SPNS2, ZRSR2, STK35, SLC29A4, SNTB1, CYP8B1, ZC3H12D, ZNF837
60.0-62.0 kDa	371	0	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	DCLRE1B, ZNF454, ZNF350, OTOA, CSPG5, OIT3, RBM46, KRT6C, UGT1A4, SLC22A7, KPNA6, GATA6, FMO3, ZNF215, RPS6KL1, KRT6A, SLC26A10P, KRT6B
62.0-64.0 kDa	324	0	3	Actin family; Apoptosis caspase/BCL; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	UTP18, EOGT, SRPK3, CLCC1, DNAJC21, GTF2H1, SLC15A4, ACOT12, C1orf87, CATSPER2, TDRKH, SPTLC3, MLLT1, ITPRIP [Actin family], KRT9, ZNF155, GNS, ZNF852
64.0-66.0 kDa	303	0	6	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	CCDC82, MFSD6L, PRKAA1, ZNF382, FAM151A, ARSI, ZNF34, GALNT13, SHC3, PRDM14, TRIM23, LILRB5, PAK4, ARID5A, C2orf42, ZBTB46, RBM47, IKZF4
66.0-68.0 kDa	259	2	4	Actin family; Cell-cycle CDK/cyclin; ER markers; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	CKAP4, ZNF583, TNFRSF11A, IGFALS, KRT1, , NOP56, MBD4, FRRS1, FARSB, FSIP1 [Actin family], IGF2BP2, CHRM3, ENC1, SLC13A1, TRIP4, MPP3, ACSM3 [Mitochondrial markers]
68.0-70.0 kDa	251	0	8	Actin family; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	high	, PES1, SLC6A13, FPGT, PTPN11, PGBD2, PIAS3, PTPN9, SUSD5, PRODH [Mitochondrial markers], KLHL41, LRIT1, KARS1, GATAD2A, POF1B, LINGO2 [Actin family], MSLN, WDR46
70.0-72.0 kDa	265	2	4	Actin family; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	RUFY2, HSPA2 [Heat shock proteins], PABPC3, F2, CLGN, SLCO2A1, ZP1, HSPA1A [Heat shock proteins], HSPA1B [Heat shock proteins], NELFB, FNBP1L, NTN4, ZSWIM9, ENOX2, FBXL4, GLCE, MOCS1, F11
72.0-74.0 kDa	219	1	3	Actin family; ER markers; Heat shock proteins; JAK-STAT; Mitochondrial markers	elevated	ZCWPW1, ANKS3, FRMD6, THEMIS2, NBPF4, SLC27A4, SYK, GLB1L2, ASNSD1, TMCC1, WDR26, WRNIP1, ZNF528, LY9, CREBRF, SLC6A14, APLP1, LIMD1
74.0-76.0 kDa	223	1	2	Actin family; Heat shock proteins; Lamin family; Mitochondrial markers; Tubulin family	elevated	SLC5A11, ERVK-24, FLRT2, TBR1, PATZ1, CHML, SLC5A9, FIGNL1, PADI4, FLRT1, RRN3, SYN1, UBASH3A, BRD7, LMNA [Lamin family], HDC, LOC122539214, SLCO1A2
76.0-78.0 kDa	201	0	6	AKT pathway; Actin family; MAPK cascade; NF-kB/Rel; Tubulin family	elevated	LNX2, CPEB3, SYTL4, FAM200B, GPCPD1, GRAMD1C, PLPPR3, CNGA2, RALBP1, USHBP1, GLB1, CCDC81, TUBGCP4 [Tubulin family], ZNF630, CLPTM1, IFT70B, BEST3, AKAP8

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
78.0-80.0 kDa	210	0	5	AKT pathway; Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	BMX [AKT pathway], SETMAR [Histone family], MARCHF7, DVL3, DLL1, NFKBIZ [NF-kB/Rel], RNF6, RGL3, TBC1D14, UVRAG, ZNF429, ZBED1, MED25, ZNF573, LTF, PHACTR4 [Actin family], PJA2, FNDC7
80.0-82.0 kDa	202	0	4	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	elevated	RNPEPL1, SKI, ZNF770, DNAAF1, KIF3A, RAP1GAP2, PCCA [Mitochondrial markers], ARHGEF16, FLYWCH1, TRAP1 [Heat shock proteins; Mitochondrial markers], C1R, SLC44A2, SLC9A7, ZNF441, KLHL11, CPEB4, PEG10, PAMR1
82.0-84.0 kDa	204	1	7	Actin family; Apoptosis caspase/BCL; Heat shock proteins; MAPK cascade; Mitochondrial markers	elevated	SANBR, VWA2, ZBTB1, MZF1, ARHGAP28, FYB2, MX2, EEF2K, FIGN, LRRFIP2 [Actin family], DZANK1, SLC6A16, VPS52, DCLK1, INTS10, MTSS1, ORC3, EXT2
84.0-86.0 kDa	150	2	3	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Mitochondrial markers; NF-kB/Rel	elevated	SOX5, SLC30A5, SLC7A14, MAP7, PIGQ, RIN1, HMMR, NIBAN2, ADAM10, PCSK1, MFN1, ARMH4, POMT2, PIK3R6, ARHGAP24, SNRK, FCHSD2, FOXM1
86.0-88.0 kDa	183	0	4	Actin family; Cell-cycle CDK/cyclin; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	HCN3, VPS51, DGCR8, LRCH3, SMURF1, STRN, MTMR12, SEC23A, MAGED1, PDZD4, ZNF184, HGS, SMURF2, DCST2, ZNF700, IL17RC, ZNF711, PCSK7
88.0-90.0 kDa	149	0	4	Actin family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	ACAP2, IFT80, , ITGB5, BRD2, LARGE1, ZNF585B, STAT3 [JAK-STAT], CDH18, CTAGE8, UNK, MFSD6, PTPN12, BRAT1, CMTR2, HECTD2, LENG8, ZER1
90.0-92.0 kDa	136	0	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; JAK-STAT; MAPK cascade; Nuclear receptor PPAR	elevated	TMPRSS6, ARHGEF7, RASAL1, WDR27, TMC6, RRM1, RSBN1, CATSPER1, KIFC2, DACT1, ADAMTS4, SUSD2, LRRC8E, CAGE1, SART1, ZNF616, TLR1, DLG3
92.0-94.0 kDa	140	0	6	Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	NLGN4Y, FGFR2, AVIL, ZNF227, SORT1, KDM1B [Histone family], ANAPC4, TMEM63A, RBM5, CSF3R, CNGB3, FAM171B, ANKS6, ELAC2, CDH17, ARHGAP26, DDX20, CDC5L
94.0-96.0 kDa	126	1	4	Actin family; Apoptosis caspase/BCL; Heat shock proteins; JAK-STAT; MAPK cascade	elevated	ANKRD20A1, , ANKRD20A2P, DDX31, COG3, EXO1, USP6NL, ANKRD20A3P, ZNF160, ZNF43, TMTC2, SLC39A10, STAT6 [JAK-STAT], TPCN1, ANKRD20A4P, E2F8, TTC7B, LRRC8A
96.0-98.0 kDa	139	0	8	Actin family; Heat shock proteins; JAK-STAT; Mitochondrial markers; NF-kB/Rel	elevated	GRIPAP1, PDCD6IP [Actin family], VWA7, VLDLR, SLC12A9, IL17RA, AHR, SEMA4D, TTC7A, ARHGAP12, NOM1, SPICE1, DCAF6, DDX24, PARP9, NLGN1, PTCHD4, ATP6V0A4
98.0-100.0 kDa	109	0	1	Actin family; Histone family; JAK-STAT; Mitochondrial markers; p53 pathway	elevated	LRRD1, DNM2, ATP6V0A2, ZHX1, MOCOS, SFMBT1, PDE4A, SKIDA1, SLC4A11, MCM10, LRRC8D, ZNF484, SP140, DPP9, TRPV4, TTC16, VAV1, PDE6B
100.0-102.0 kDa	141	0	5	Actin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	XAB2, ESYT3, ZCCHC14, CIZ1 [Actin family], CTNNA1, ENPP3, NFATC2, RIN2, C2orf78, ZNF473, RBM25, PSMD2, CLCA1, SF3B2, ERMP1, KIF20A, CDH4, CHSY3
102.0-104.0 kDa	127	0	3	Actin family; Apoptosis caspase/BCL; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	elevated	USP20, CCDC178, ZZZ3, CHRD, SLFN13, PCDHA5, PCDHA2, AFTPH, TCAF1, AASS [Mitochondrial markers], CRACDL, KSR1, KCNQ5, CIP2A, MTBP, PRRT3, GRM7, KIF18A
104.0-106.0 kDa	118	0	2	Actin family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	TMTC3, GRIK3, NPIPB12 [Actin family], MAP3K14 [MAPK cascade], USP26, PEX6, CC2D1A, EXOC2, ADGRD2, GOLGA6L25, ANKLE2, , USP29, AGBL2, THBS3, TNPO3, OCRL, ROR1
106.0-108.0 kDa	102	0	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Mitochondrial markers	elevated	DDX12P, TRAK1, DLGAP3, GTF2IRD1, CLSTN3, WDR3, BCLAF1 [Apoptosis caspase/BCL], RB1, NCOA7, PHC3, ANO6, GALNT5, CCDC14, OSBPL6, TRPC6, MORC4, NUP107, ZC3H18
108.0-110.0 kDa	92	0	1	Actin family; Cell-cycle CDK/cyclin; Lamin family; MAPK cascade	elevated	DLGAP4, MAML1, HERVK_113, ERVK-19, EPHA1, ESPNL, CTNND1, INTS4, PIGG [Lamin family], CCDC80, IPO13, EIF4ENIF1, SLC4A9, KIF28P, EPHA2, EPHB4, MAP3K13 [MAPK cascade], DDX11
110.0-112.0 kDa	86	0	1	Actin family; Histone family; Mitochondrial markers	elevated	ICE2, RAI14, KIF23, TMEM132A, TAF4, EFCAB13, EPHA3, MIB1, ZSWIM4, USP37, SNX14, VPS18, MERTK, ATP2A1, FNBP4, MED24, SAP130 [Histone family], EPHB3
112.0-114.0 kDa	93	1	3	Actin family; Apoptosis caspase/BCL; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR	elevated	PKN2, MLLT6, EPHA7, TRPC4, GRID1, STK10, SIPA1, SNX13, MAP1S, SYNE3, SALL4, KANSL1L, ATP1A2, NAA25, TOP3A, GTF2I, USP15, SEPTIN4
114.0-116.0 kDa	71	0	1	Actin family; Apoptosis caspase/BCL; Mitochondrial markers	elevated	TMPRSS9, PPFIBP1, ANO1, ATP4A, ATP1A4, SYCP1, PUM2, FAN1, NCAPG, SEMA6A, OGDHL [Mitochondrial markers], ITGA9, UBAP2L, ITGA5, TNK2, GIGYF1 [Actin family], ANO3, NLRC3
116.0-118.0 kDa	88	0	2	Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers	elevated	AGBL3, GARRE1, RIMBP2, PSD3, ITGAV, TAOK1, PCDH7, TMEM132E, NLRC4, KL, MYO1D, HIP1 [Actin family], DGKI, PSD4, ADGRF3, EPHA6, DCLRE1A, RASA1
118.0-120.0 kDa	73	0	3	AKT pathway; Actin family; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	AXDND1, FRMD4B, INTS3, AMOT, RBM12B, MRTFB, NLRP3, ANKRD18B, U2SURP, TUBGCP5 [Tubulin family], SEC24C, PTK7, MYO1A, VARS2 [Mitochondrial markers], RIPOR2 [Actin family], HERC4, TSHZ3, ARHGAP30
120.0-122.0 kDa	72	0	2	Actin family; Histone family; MAPK cascade; p53 pathway	elevated	GEMIN4, GUCY2D, LATS2, NLRP12, SUGP2, STK11IP [Actin family], LUZP1, ADCY7, NUTM1, DNA2, USPL1, NLRP2, RAPGEF1, FAM120C, SEMA5A, SLC12A4, KDM4A, CASR
122.0-124.0 kDa	71	1	1	AKT pathway; Actin family	elevated	ANKIB1, TTI1 [Actin family], PALM2AKAP2, MUC1, FNIP2 [Actin family], TRHDE, CEP131, MEGF10, USP25, CP, ZNF658, MAML3, DSG2, TMEM132D, MYT1, BNC2, PTPRH, BUB1 [AKT pathway]
124.0-126.0 kDa	70	0	3	Actin family; JAK-STAT; p53 pathway	elevated	AMER1, C6orf132, OTUD4, ZEB1, SORBS2, HELQ, CCDC73, PLD1, ANKRD24, TBC1D31, ANLN, SRGAP1, PIK3CA, ZNF516, RET, PDE3B, UPF1, RANBP17
126.0-128.0 kDa	61	0	2	Actin family	elevated	ALG13, SENP6, PCDH18, SLC12A5, PCDH17, PCDH19, SLC4A5, CILP2, SMC6, BTBD7, PIK3CG, NPIPB5 [Actin family], PUM1, DENND2B, FHOD1, FBLN2, IGSF9, KIAA1614
128.0-130.0 kDa	70	0	4	Actin family; Lamin family; Mitochondrial markers	elevated	CEP128, MYBPC2, ERC1 [Actin family], MCF2L, DHX34, SORCS2, NCKAP1L, ZFPM2, POLR1B, RFC1, MYH16, MYBPC1, USP7, ABL2, RBL2, ANKFY1, INPP5F, TRAPPC9
130.0-132.0 kDa	57	0	1	Actin family; JAK-STAT; Lamin family	elevated	SPOCD1, CCDC40, PRDM10, XPO4, AP3D1, ITGAE, STYXL2 [Actin family], PPP2R3A, ZBED4, FILIP1L [Actin family; Lamin family], HEPH, KCNT2, SPATA31C1, MAN2A2, PTCH2, FNIP1 [Actin family], ADCY6, ZSWIM5
132.0-134.0 kDa	73	0	1	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Tubulin family	elevated	TDRD1, TRERF1, GRID2IP, DNAAF9, RIPOR1 [Actin family], RGS3, GRM1, EHMT2 [Histone family], ARID5B, GRM5, LEPR, PARD3B, CILP, TDRD12, TBCD [Tubulin family], ARHGAP20, CTNND2, NRDE2
134.0-136.0 kDa	65	0	3	NF-kB/Rel	elevated	VWA3A, PLCB2, CCDC136, CDAN1, NOMO3, ATP11B, ATP2B3, GRIN2C, IGDCC4, ZBTB38, RBM20, CRB2, EGFR, WASHC5, TRPM4, VCPIP1, NOMO1, NLRP5
136.0-138.0 kDa	46	0	0	Actin family; Tubulin family	routine	MN1, ANO8, SLC4A7, PCDH9, ADAR, GSE1, ADAMTS16, PER1, XPO5, CAND1, NID1, ATF7IP [Actin family], WASHC4, ZEB2, PER2, WWC3, SASH1, WDR11
138.0-140.0 kDa	49	0	0	Actin family; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	routine	ATP13A3, ANKS1B, FILIP1 [Actin family; Lamin family], MMRN1, ZBTB40, MECOM [Histone family], REV1, TAOK2, KCNT1, PTPRO, COPA, TPP2, ITPRID2, PHKA2, AGTPBP1, EDRF1, COL3A1, PLCB1
140.0-142.0 kDa	51	0	1	Actin family; Histone family	routine	L1CAM, EHBP1, KIF4B, ADCY8, USP40, PRDM16 [Histone family], ZC3H4, ATXN2, ARHGEF10L, SALL1, PIPP5K2, CNTNAP3B [Actin family], VARS1, DHX38, SCAF8, TBC1D9B, MTR, SYNRG
142.0-144.0 kDa	40	0	1	Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Tubulin family	elevated	TNKS, ARHGAP29, MAP4K4 [MAPK cascade], PHLDB2, NPC1, TRAPPC10, CACHD1, MMS22L, ULK4, AMBRA1, SORBS1, MAP3K6 [MAPK cascade], ADGRA2, SLK, TTBK1 [Tubulin family], ARID4A, BCR, NEK1
144.0-146.0 kDa	39	0	1	Actin family	routine	TNN, RTL9, FRMPD2, UMODL1, FRMPD4, STAG1, IARS1, AATK, ZNF423, PFAS, SGO2, FLII, TBC1D32, AFF2, SH3TC2, SIK3, ZNF335, COL5A2
146.0-148.0 kDa	41	0	2	Actin family; MAPK cascade	elevated	UBN2, ADGRA3, UBE4B, SPAG9 [Actin family], ABCB11, XDH, TBC1D4, USP31, CLUH, RPGRIP1 [Actin family], PHLPP2, ATP8B3, PCDH11Y, ERBB4, SH3TC1, IGSF9B, DENND5A, WASHC2C
148.0-150.0 kDa	40	0	1	Actin family; Apoptosis caspase/BCL; Histone family	routine	ZMYM6, SPATA31F1, ERBB3, CNTNAP2 [Actin family], ROBO3, ASTN2 [Actin family], SYNGAP1, JCAD, CYFIP2 [Actin family], RNF123, PLCG1, SPATA31A1, SPATA31A5, SPATA31A3, NPC1L1, SPATA31A7, MYBBP1A, PITPNM2
150.0-152.0 kDa	41	0	1	Actin family; MAPK cascade	routine	NFASC, GIGYF2 [Actin family], MTUS2, CARMIL3, ELP1, SBNO2, COL17A1, IBTK, MAP3K19 [MAPK cascade], PALLD, KIF7, ADCY9, FLT1, YEATS2, PDS5A, NPHP3, TONSL, TTC21B
152.0-154.0 kDa	34	0	1	Actin family; Histone family	routine	DISP2, KIF16B, TCOF1, SNED1, BRD4, MST1R, NSD2 [Histone family], ABCC12, ZMYM3, RRBP1, SOS1, ZNF804B, TNS2, KDM2B, ATRNL1 [Actin family], RPAP1, FLT4, MSH6
154.0-156.0 kDa	30	0	1	Actin family; MAPK cascade	routine	KDM6A, CRB1, NPAT, ABCC11, SBNO1, CEMIP2, MAP3K5 [MAPK cascade], PLCH2, CARMIL2, KIAA0232, IGF1R, PRX, ZMYM2, TNIK [Actin family], RTL1, AKNA, NUP155, DHX29
156.0-158.0 kDa	25	0	2	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	elevated	RPRD2, CNTNAP1 [Actin family], KTN1, INSR, NINL, RGS12, ZFYVE9, MED23, CEFIP [Actin family], NCOA1, GPRASP1, ARHGAP31, BCL9L [Apoptosis caspase/BCL], SPATA31E1, NCAPD2, ATP7B, USP47, RFX7
158.0-160.0 kDa	25	0	1	Actin family; Cell-cycle CDK/cyclin	routine	ANKRD30B, ROCK1, SSH2, SCAPER [Cell-cycle CDK/cyclin], ERBIN, DCC, BICRA [Actin family], ATRN [Actin family], ATAD2, FHOD3, USP6, MAGI2, ANKRD30A, BLM, NCOA2, CD163L1, RIC1, DROSHA
160.0-162.0 kDa	37	0	1	Histone family	routine	NEO1, DAPK1, KIF15, YTHDC2, ATP10D, KIF27, RIMS2, SMG6, C1orf167, PTCH1, MED14, COL4A1, ABCC5, FGD6, CPSF1, ROCK2, NOS1, TRAPPC8
162.0-164.0 kDa	33	0	1	MAPK cascade	routine	PTPRG, ANKAR, PTPRK, NUP160, PTPRT, MYOM3, ARAP1, ARHGAP23, ZNF608, CLIP1, PTPRU, WRN, EEA1, UACA, HFM1, ADGRL1, RAD54L2, FANCA
164.0-166.0 kDa	39	0	3	Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers	elevated	COL4A4, LOC107987545, FANCD2, CDK12 [Cell-cycle CDK/cyclin], CUX1, GPATCH8, BLTP3B, CECR2, CEP164, MAP3K1 [MAPK cascade], MAGI1, SCAF11, ARHGEF40, PDS5B, CCDC88B, DOT1L [Histone family], SHROOM4, MYOM2
166.0-168.0 kDa	25	0	0		routine	MRC1, GRIN2B, FAM193A, EIF3A, NISCH, MRC2, ZNF518A, CC2D2B, RALGAPB, NEURL4, LRRC9, KCP, FYCO1, C4orf50, GOLGA3, RAPGEF2, NEXMIF, COL4A2
168.0-170.0 kDa	21	0	0		routine	CAMSAP2, CFTR, PLEKHH2, AHDC1, ERCC6, ERICH3, MED1, GEMIN5, PLA2R1, NCAPD3, ZFYVE16, CCDC18, TULP4, DCAF1, KIAA0586, ABCC3, CLASP1, FAM135A
170.0-172.0 kDa	21	0	1	AKT pathway; Actin family; Lamin family	routine	DIP2A [Actin family], ARHGAP35, URB2, EPRS1, DLC1, NHSL1, MAST1, TOPBP1, PIK3C2G, DIP2C [Actin family], BAZ1B [AKT pathway], DISP1, TRPM2, LAMC3 [Lamin family], TUT7, AQR, DIP2B [Actin family], ADGRB3
172.0-174.0 kDa	16	0	0		routine	COL5A3, RERE, CDC42BPG, ARHGAP5, ADGRB2, ZMYM4, SYNM, EFCAB6, PCF11, SYNJ1, ARHGEF12, EFCAB5, LTBP4, FRMPD1, MUC22, ADGRB1
174.0-176.0 kDa	21	0	0		routine	KDM5D, ABCC2, ABCC9, TOP2A, NWD1, UGGT2, AGL, SETBP1, CCDC141, CEP170, DUOX2, SKIC3, EIF4G1, COL24A1, SPATA31D1, SYCP2, KDM5B, KDM5C
176.0-178.0 kDa	22	0	0	Lamin family; Nuclear receptor PPAR	routine	SHROOM2, ALK, KDM6B, EIF4G3, INO80, VWDE, ARHGEF5, ABCC8, ERCC6L2, LRRC7, UGGT1, DUOX1, DNMBP, NES, TIAM1, PPRC1 [Nuclear receptor PPAR], LAMC1 [Lamin family], ADAMTS12
178.0-180.0 kDa	14	0	0	Actin family	routine	COL18A1, PLEKHG4B, NRK, CFAP74, PHRF1, WIZ, BAZ1A, PTPN23, NHS [Actin family], LRP5, THSD7B, TMEM131L, RAPGEF6, HECW1

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
180.0-182.0 kDa	27	0	1	Actin family; MAPK cascade	routine	FMN2, WDR97, LRP6, IQGAP2, NRXN3, LRRC37A3, ALPK3, UNC13B, RIMBP3, MROH2B, QRICH2, PEG3, KATNIP [Actin family], ROBO1, RIMBP3C, RIMBP3B, PAPPA, CTTNBP2 [Actin family]
182.0-184.0 kDa	14	0	1		routine	TRPM1, PREX2, SIGLEC1, KIF21B, THOC2, RB1CC1, DNMT1, TOP2B, MADD, COL5A1, ALS2, CAMTA1, ABCA8, MTCL2
184.0-186.0 kDa	14	0	1	Actin family	routine	ADAMTS7, SAMD9, ABCA6, ABCA9, SAMD9L, RNF17, SMARCA4 [Actin family], PHLPP1, IQGAP3, PIK3C2B, NRXN2, TUT4, THSD7A, CROCC2
186.0-188.0 kDa	19	0	0	Histone family	routine	SETD1A [Histone family], CC2D2A, PREX1, MYO3A, SPHKAP, KIF14, ABCA5, LTBP1, COL27A1, EIF2AK4, KIAA1210, CLTCL1, C3, ADCY10, KIF21A, USP54, WDR90, MYOM1
188.0-190.0 kDa	13	0	0		routine	LRRC37A, C5, LRRC37A2, ADAMTSL3, PRDM2, TASOR, RIMS1, PLCH1, IQGAP1, TOGARAM1, MROH2A, ADGB, QSER1
190.0-192.0 kDa	19	0	1		routine	CASZ1, C4orf54, TIAM2, NBPF26, MON2, SIPA1L2, BCORL1, PIK3C2A, TOPAZ1, KIF1A, CUL7, SHANK3, ABCA3, KNDC1, AKAP12, KDM3B, CLTC, ARHGEF28
192.0-194.0 kDa	21	0	0	Lamin family	routine	KDM5A, BCOR, RICTOR, STRCP1, PARP4, LMO7, C4B, C4A, MARF1, PBRM1, STRC, UNC13A, SHPRH, PEAK1, ADAMTSL1, ARAP2, ITSN2, SCN7A
194.0-196.0 kDa	19	0	0	Lamin family	routine	TNRC6B, XRN1, CDC42BPB, KIF26A, SIPA1L3, POLR1A, RESF1, LTBP2, ITSN1, TJP1, LOC124905564, LOC102724250, CEP152, FRMPD3, DLEC1, LOC124905558, PKD1L3, GCC2
196.0-198.0 kDa	24	0	0	Actin family	routine	NUP188, PATJ, ANKRD26, MAST2, KIDINS220 [Actin family], ZC3H13, RGPD1, CHD1, KIAA1671, TNS1, NRAP, RGPD4, ZSWIM8, CDC42BPA, RGPD2, CCDC180, ARID2, NWD2
198.0-200.0 kDa	14	0	0	Actin family; Lamin family	routine	LAMB1 [Lamin family], LY75, WNK3, PAPPA2, XIRP1 [Actin family], N4BP2, RGPD5, RGPD8, KIAA1549L, BTBD8, SUPT6H, LRRIQ1, MAP2, ANKRD36C
200.0-202.0 kDa	15	0	1	Tubulin family	routine	SLX4, SIPA1L1, TSPOAP1, FOCAD, UBR1, NALCN, TUBGCP6 [Tubulin family], UBR2, LTN1, DCDC1, TSC2, ZNF646, SHANK2, RBBP6, TNRC6C
202.0-204.0 kDa	13	0	0	Lamin family	routine	VIRMA, ARFGEF2, ITGB4, TANC1, KIF13A, NAV1, LAMA4 [Lamin family], KIF13B, PARP14, MYO5C, RAI1, BRWD3, ZNF236
204.0-206.0 kDa	11	0	0	Histone family; p53 pathway	routine	ECPAS, KIF1B, NLRC5, PPL, SCN11A, KMT2E [Histone family], NUP210 [p53 pathway], PLXNB2, TMEM131, NACA, FNDC1
206.0-208.0 kDa	14	0	0	Actin family	routine	MYO16, ANK1, GBF1, PHIP [Actin family], CPAMD8, AFDN, PLXNB3, BTAF1, PCSK5, OBSL1, TAF1L, ATAD5, PLXNA3, BRCA1
208.0-210.0 kDa	16	0	0		routine	SCN4A, NYNRIN, SBF1, SBF2, NCKAP5, PDCD11, HCFC1, ARFGEF1, ZNF106, CATSPERT, DENND4A, FER1L6, SI, MTCL1, NOTCH4, SPEF2
210.0-212.0 kDa	18	0	0	p53 pathway	routine	TNRC6A, AKAP11, NUP210L [p53 pathway], KIF20B, MYLK, KIAA1549, RALGAPA2, ANKRD31, TICRR, PRR12, PLXNA1, PLXNA2, BAZ2A, PSME4, CHD2, WDR81, TCF20, DOCK2
212.0-214.0 kDa	16	0	1	Histone family; p53 pathway	routine	PLXND1, LRP4, CACNA1S, UBR3, PLXNA4, TRPM7, DENND4C, SETD1B [Histone family], ATG2A, PTPRF, TP53BP1 [p53 pathway], NUP214 [p53 pathway], MYO5B, MIA3, CFAP44, DLG5
214.0-216.0 kDa	11	0	1	Cell-cycle CDK/cyclin	routine	KIAA1217, GREB1L, ANKRD36, TAF1, ADAMTS20, PTPRD, VWA8, CDK5RAP2 [Cell-cycle CDK/cyclin], DOCK5, DOCK1, MYO5A
216.0-218.0 kDa	14	0	0	p53 pathway	routine	CCDC88A, PCDH15, GREB1, ADAMTS9, ANAPC1, SHROOM3, PTPRS, POLR2A, CFAP65, AGRN, ARHGAP21 [p53 pathway], PWWP4, EML6, RLF
218.0-220.0 kDa	10	0	0	Actin family	routine	CHD4, MCM3AP, LCT, DICER1, HELZ, MICAL2 [Actin family], NCOA6, EML5, THADA, TANC2
220.0-222.0 kDa	8	0	0		routine	ZNFX1, ZNF638, SCN10A, CACNA1F, AK9, MPDZ, ARHGEF17, TDRD15
222.0-224.0 kDa	18	0	1	Apoptosis caspase/BCL	routine	HEATR5A, PCNX3, MYH15, DSCAM, EXPH5, CASP8AP2 [Apoptosis caspase/BCL], MYH8, MYH2, CHD5, MYH4, MYH7, MYH1, MYH13, CR1, MYH6, TET2, KIF26B, MYH3
224.0-226.0 kDa	12	0	0	Actin family; Histone family	routine	MICAL3 [Actin family], PTPRB, HEATR5B, SPATA31H1, DSCAML1, SHANK1, KAT6A [Histone family], DOCK4, SCN8A, LRRK1, CKAP5, MYH7B
226.0-228.0 kDa	15	0	0	Actin family	routine	SCN3A, ZFC3H1, SCN9A, SMCHD1, MYH9, LPA, CHD3, MDC1, OTOF, SCN5A, MYH11, TRIP11 [Actin family], MYH14, NUP205, SCN2A
228.0-230.0 kDa	12	0	0		routine	BRD10, CCDC88C, CROCC, TRIP12, PCM1, PRRC2A, SCN1A, MYH10, PHF3, DOCK6, RALGAPA1, TTC3
230.0-232.0 kDa	6	0	0	Actin family; Histone family	routine	ARHGAP32, KAT6B [Histone family], CIT [Actin family], EVPL, TRPM6, CRYBG1
232.0-234.0 kDa	6	0	0		routine	FANCM, PLXNB1, ATG2B, DOCK3, MYO18A, ESPL1
234.0-236.0 kDa	6	0	0		routine	ABCA7, MYOF, TET1, ANKRD12, LOXHD1, ARMCX4
236.0-238.0 kDa	12	0	0		routine	DOCK9, TDRD6, ZGRF1, PI4KA, ALPK2, PIKFYVE, PCNX2, DYSF, PRR14L, MYO10, DOCK11, FER1L5
238.0-240.0 kDa	7	0	0		routine	FREM3, NUMA1, DOCK8, GTF3C1, MED13, SDK2, TECTA
240.0-242.0 kDa	10	0	0		routine	MED12L, BAZ2B, ARFGEF3, RP1, TNC, MALRD1, MYO7B, YLPM1, USF3, ASXL3
242.0-244.0 kDa	15	0	0		routine	ARID1A, SDK1, HEATR1, DOCK7, MED13L, WNK2, PRRC2B, CAD, MED12, NIN, MYO9B, NOTCH3, DIDO1, ARID1B, APC2
244.0-246.0 kDa	5	0	0		routine	FREM1, INTS1, SNRNP200, CACNA1I, CACNA1D
246.0-248.0 kDa	6	0	0		routine	CABIN1, SPTB, COL6A6, ZNF407, SVIL, ICE1
248.0-250.0 kDa	6	0	0		routine	FLG2, SORL1, GON4L, RTTN, CACNA1C, DOCK10
250.0-252.0 kDa	9	0	0		routine	CSPG4, KNTC1, WNK1, UNC13C, ZNF318, ASCC3, F5, SPAG17, SEC16A
252.0-254.0 kDa	4	0	0		routine	RP1L1, AHCTF1, BLTP2, TCHH
254.0-256.0 kDa	8	0	0		routine	ABCA1, URB1, MYO7A, DNAJC13, PTPRZ1, PKDREJ, NAV3, ABCA4
256.0-258.0 kDa	3	0	0		routine	AKAP6, MUC6, GPR179
258.0-260.0 kDa	6	0	0		routine	CIC, DOP1B, PCNX1, PLCE1, CACNA1H, HIVEP3
260.0-262.0 kDa	8	0	0	Actin family	routine	C2CD3, DMBT1, PTPRQ, GOLGA4, ACAN, TRIOBP [Actin family], POLE, CACNA1E
262.0-264.0 kDa	6	0	0		routine	CACNA1G, CACNA1B, FAM186A, BRWD1, SON, ROS1
264.0-266.0 kDa	8	0	0	Histone family	routine	EP300 [Histone family], OTOGL, PDE4DIP, CREBBP, KNL1, NOTCH2, ACACA, ZDBF2
266.0-268.0 kDa	3	0	0		routine	CNOT1, F8, TPR
268.0-270.0 kDa	8	0	0		routine	NAV2, NBAS, TASOR2, CNTRL, HIVEP2, ANKHD1, TLN1, ABCA2
270.0-272.0 kDa	5	0	0		routine	NCOR1, MAP1B, TTC28, SPTBN2, TLN2
272.0-274.0 kDa	6	0	0		routine	FN1, NOTCH1, PKD1L2, FASN, PRPF8, NCOR2
274.0-276.0 kDa	6	0	0		routine	ZFHX2, ANKRD17, IGF2R, RIF1, SPTBN1, STAB1
276.0-278.0 kDa	5	0	0		routine	ACACB, PTPN13, STAB2, DOP1A, MGAM2
278.0-280.0 kDa	4	0	0	Lamin family	routine	FLNB [Lamin family], SPG11, GVINP1, CEP192
280.0-282.0 kDa	5	0	0	Lamin family	routine	SPTA1, BAHCC1, FLNA [Lamin family], CEP250, CUL9
282.0-284.0 kDa	3	0	0		routine	HRNR, CACNA1A, ATRX
284.0-286.0 kDa	6	0	0		routine	MAST4, JMJD1C, SPTAN1, ZFYVE26, ZNF462, MYO18B
286.0-288.0 kDa	3	0	0	Histone family	routine	LRRK2, PIEZO1, SETD2 [Histone family]
288.0-290.0 kDa	5	0	1		routine	MTOR, SPTBN4, HECTD1, POLQ, COL6A5
290.0-292.0 kDa	7	0	0	Lamin family	routine	CEP290, USP9X, TEP1, CHD8, IGSF10, FLNC [Lamin family], USP9Y
292.0-294.0 kDa	6	0	0	Histone family	routine	EPG5, MYO9A, GCN1, ABCA12, KMT2B [Histone family], BDP1
294.0-296.0 kDa	4	0	0		routine	USP24, CEP295, COL7A1, UNC79
296.0-298.0 kDa	3	0	0	Histone family	routine	NSD1 [Histone family], HIVEP1, ANKRD11
298.0-300.0 kDa	1	0	0		routine	SRRM2
300.0-302.0 kDa	4	0	0		routine	FBN3, TENM3, ATR, PDZD2
302.0-304.0 kDa	4	0	0		routine	NBEAL2, SETX, MEGF8, CFAP46
304.0-306.0 kDa	7	0	0		routine	ITPR3, TG, ZNF292, TENM1, CHD6, MAP1A, ZAN
306.0-308.0 kDa	4	0	0		routine	NBEAL1, AKAP13, TENM2, TENM4
308.0-310.0 kDa	4	0	0		routine	ITPR2, VWF, UBR5, TACC2
310.0-312.0 kDa	1	0	0		routine	APC
312.0-314.0 kDa	4	0	0		routine	MGAM, MXRA5, FBN1, ITPR1
314.0-316.0 kDa	5	0	0		routine	TNRC18, FBN2, OTOG, TEX15, PKD1L1
316.0-318.0 kDa	4	0	0		routine	NIPBL, CENPE, PRRC2C, CELSR2

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
318.0-320.0 kDa	4	0	0		routine	PIEZO2, UTP20, LRBA, NF1
322.0-324.0 kDa	1	0	0		routine	HELZ2
326.0-328.0 kDa	2	0	0		routine	CHD9, NBEA
328.0-330.0 kDa	1	0	0		routine	CELSR1
330.0-332.0 kDa	4	0	0		routine	BOD1L1, CRYBG3, ZZEF1, DSP
332.0-334.0 kDa	4	0	0	Histone family	routine	ASH1L [Histone family], COL12A1, WDR87, ADGRG4
334.0-336.0 kDa	1	0	0		routine	CHD7
336.0-338.0 kDa	4	0	0	Lamin family	routine	MGA, TRANK1, LAMA1 [Lamin family], DMXL1
338.0-340.0 kDa	5	0	0		routine	MYO15B, BPTF, FRY, FRYL, DMXL2
340.0-342.0 kDa	2	0	0		routine	KALRN, PRUNE2
342.0-344.0 kDa	5	0	1	Lamin family	routine	NBPF14, EP400, SRCAP, COL6A3, LAMA2 [Lamin family]
344.0-346.0 kDa	1	0	0		routine	MUC3A
346.0-348.0 kDa	3	0	0		routine	DCHS1, TRIO, HTT
350.0-352.0 kDa	5	0	0		routine	ATM, EYS, CEP350, FREM2, CFAP54
352.0-354.0 kDa	2	0	0		routine	REV3L, WDFY4
354.0-356.0 kDa	1	0	0		routine	SPEG
356.0-358.0 kDa	2	0	0		routine	DNAH12, CENPF
358.0-360.0 kDa	3	0	0		routine	CELSR3, RANBP2, MKI67
360.0-362.0 kDa	3	0	0		routine	VPS13A, CFAP47, CPLANE1
362.0-364.0 kDa	2	0	0		routine	APOLTP, UNC80
366.0-368.0 kDa	1	0	0	Lamin family	routine	LAMA3 [Lamin family]
368.0-370.0 kDa	1	0	0		routine	CDH23
370.0-372.0 kDa	1	0	0		routine	DCHS2
372.0-374.0 kDa	1	0	0		routine	VCAN
376.0-378.0 kDa	1	0	0		routine	GOLGB1
378.0-380.0 kDa	2	0	0		routine	SZT2, PCNT
380.0-382.0 kDa	1	0	0		routine	CSMD2
382.0-384.0 kDa	1	0	0	Actin family	routine	XIRP2 [Actin family]
384.0-386.0 kDa	1	0	0		routine	BRCA2
388.0-390.0 kDa	2	0	1	NF-kB/Rel	routine	RELN [NF-kB/Rel], CSMD1
390.0-392.0 kDa	1	0	0		routine	SVEP1
392.0-394.0 kDa	1	0	0		routine	ZFHX4
394.0-396.0 kDa	3	0	0		routine	UTRN, WDFY3, MYO15A
398.0-400.0 kDa	3	0	0	Lamin family	routine	CUBN, LAMA5 [Lamin family], DNAH14
402.0-404.0 kDa	1	0	0	Actin family	routine	SPEN [Actin family]
404.0-406.0 kDa	3	0	0		routine	USP34, ZFHX3, CSMD3
408.0-410.0 kDa	1	0	0		routine	ASPM
410.0-412.0 kDa	1	0	0		routine	SMG1
412.0-414.0 kDa	1	0	0		routine	ZNF469
416.0-418.0 kDa	2	0	0		routine	BSN, SPTBN5
422.0-424.0 kDa	1	0	0		routine	VPS13C
426.0-428.0 kDa	1	0	0		routine	DMD
428.0-430.0 kDa	1	0	0		routine	LYST
430.0-432.0 kDa	1	0	0	Histone family	routine	KMT2A [Histone family]
432.0-434.0 kDa	1	0	0		routine	ANK2
434.0-436.0 kDa	2	0	0		routine	FLG, NBPF10
436.0-438.0 kDa	1	0	0		routine	TRRAP
438.0-440.0 kDa	1	0	0		routine	HECTD4
440.0-442.0 kDa	1	0	0		routine	NBPF19
442.0-444.0 kDa	1	0	0		routine	FRAS1
446.0-448.0 kDa	1	0	0		routine	PKHD1
448.0-450.0 kDa	2	0	0		routine	VPS13B, CMYA5
450.0-452.0 kDa	1	0	0		routine	MUC17
452.0-454.0 kDa	1	0	0		routine	AKAP9
458.0-460.0 kDa	1	0	0		routine	TNXB
460.0-462.0 kDa	2	0	0		routine	ALMS1, DNAH7
462.0-464.0 kDa	1	0	0		routine	PKD1
464.0-466.0 kDa	1	0	0		routine	PKHD1L1
468.0-470.0 kDa	2	0	1	Heat shock proteins	routine	HSPG2 [Heat shock proteins], PRKDC
470.0-472.0 kDa	1	0	0		routine	DNAH3
474.0-476.0 kDa	1	0	0		routine	DNAH6
478.0-480.0 kDa	1	0	0		routine	FAT2
480.0-482.0 kDa	2	0	0		routine	ANK3, HUWE1
486.0-488.0 kDa	1	0	0		routine	DNAH1
490.0-492.0 kDa	1	0	0		routine	VPS13D
492.0-494.0 kDa	1	0	0		routine	DYNC2H1
500.0-502.0 kDa	1	0	0		routine	FAT3
504.0-506.0 kDa	1	0	0		routine	LRP1
506.0-508.0 kDa	2	0	0		routine	FAT1, DNAH2
508.0-510.0 kDa	1	0	0		routine	DNAH17
510.0-512.0 kDa	1	0	0		routine	DNAH9
512.0-514.0 kDa	1	0	0		routine	MYCBP2
514.0-516.0 kDa	4	0	0		routine	DNAH8, DNAH10, LRP1B, APOB
516.0-518.0 kDa	1	0	0		routine	STARD9
520.0-522.0 kDa	3	0	0		routine	DNAH11, SACS, LRP2
526.0-528.0 kDa	1	0	0		routine	HERC2
528.0-530.0 kDa	1	0	0		routine	DNAH5
530.0-532.0 kDa	2	0	0		routine	BIRC6, PLEC
532.0-534.0 kDa	3	0	0		routine	HERC1, DYNC1H1, DNHD1
540.0-542.0 kDa	1	0	0	Histone family	routine	KMT2C [Histone family]
542.0-544.0 kDa	2	0	0		routine	MUC4, FAT4
544.0-546.0 kDa	1	0	0		routine	HMCN2

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
546.0-548.0 kDa	1	0	0		routine	SSPOP
550.0-552.0 kDa	1	0	0		routine	MUC2
552.0-554.0 kDa	1	0	0		routine	RYR3
554.0-556.0 kDa	2	0	0		routine	BLTP1, EPPK1
558.0-560.0 kDa	1	0	0		routine	MUC12
560.0-562.0 kDa	1	0	0		routine	PCLO
564.0-566.0 kDa	2	0	0		routine	RYR2, RYR1
572.0-574.0 kDa	2	0	0		routine	FCGBP, UBR4
574.0-576.0 kDa	2	0	0		routine	USH2A, HYDIN
576.0-578.0 kDa	1	0	0		routine	ABCA13
584.0-586.0 kDa	1	0	0		routine	MUC5AC
590.0-592.0 kDa	1	0	0		routine	RNF213
592.0-594.0 kDa	1	0	0	Histone family	routine	KMT2D [Histone family]
594.0-596.0 kDa	1	0	0		routine	NBPF20
596.0-598.0 kDa	1	0	0		routine	MUC5B
612.0-614.0 kDa	1	0	0		routine	HMCN1
616.0-618.0 kDa	1	0	0		routine	AHNAK2
628.0-630.0 kDa	1	0	0		routine	AHNAK
632.0-634.0 kDa	1	0	0		routine	MDN1
692.0-694.0 kDa	1	0	0		routine	ADGRV1
780.0-782.0 kDa	1	0	0	Actin family	routine	FSIP2 [Actin family]
796.0-798.0 kDa	1	0	0		routine	SYNE2
800.0-802.0 kDa	1	0	0		routine	LRTM3
804.0-806.0 kDa	1	0	0		routine	MUC19
838.0-840.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
860.0-862.0 kDa	1	0	0		routine	DST
868.0-870.0 kDa	1	0	0		routine	OBSCN
986.0-988.0 kDa	1	0	0		routine	NEB
1010.0-1012.0 kDa	1	0	0		routine	SYNE1
1396.0-1398.0 kDa	1	0	0		routine	MUC3B
1518.0-1520.0 kDa	1	0	0		routine	MUC16
3816.0-3818.0 kDa	1	0	0		routine	TTN

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
0.26	TRDD1	P0DPR3	T cell receptor delta diversity 1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.29	TRBD1	P0DPI4	T cell receptor beta diversity 1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.44	IGHD1-1	P0DOY5	Immunoglobulin heavy diversity 1-1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.73	IGHD5-12	A0A0U1RQB9	Immunoglobulin heavy diversity 5-12 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.75	IGHD5OR15-5B	A0A0G2JLJ8	Immunoglobulin heavy diversity 5/OR15-5A (non-functional) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.82	IGHD6-19	A0A1Y8EKQ5	Immunoglobulin heavy diversity 6-19 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.85	IGHD6-13	A0A1Y8EI39	Immunoglobulin heavy diversity 6-13 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.09	IGHD3-10	A0A0J9YXN1	Immunoglobulin heavy diversity 3-10 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.10		A0ACI8PYV1	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.12	IGLJ4	A0A0A0MTA1	Immunoglobulin lambda joining 4 (non-functional) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.22	IGHD3-9	A0A0J9YW22	Immunoglobulin heavy diversity 3-9 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.25	IGKJ4	A0A0A0MT69	Immunoglobulin kappa joining 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.31	IGKJ3	A0A0A0MT96	Immunoglobulin kappa joining 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.31	IGKJ2	A0A0A0MT85	Immunoglobulin kappa joining 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.35	IGLJ5	A0A0A0MT86	Immunoglobulin lambda joining 5 (non-functional) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.35	IGHD3-3	A0A0J9YWD0	Immunoglobulin heavy diversity 3-3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.36	IGKJ5	A0A0A0MTA3	Immunoglobulin kappa joining 5 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.39	LOC128462377	A0A804HJT0	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.39	IGKJ1	A0A0A0MT89	Immunoglobulin kappa joining 1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.44	TRBJ2-2P	A0A0A0MTA2	T cell receptor beta joining 2-2P (non-functional) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.47	IGLJ7	A0A0A0MT92	Immunoglobulin lambda joining 7 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.48		A0A8I5KPI3	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.63	TRBJ1-2	A0A0J9YX06	T cell receptor beta joining 1-2	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.64	TRBJ1-1	A0A0J9YXA8	T cell receptor beta joining 1-1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.66	TRBJ2-6	A0A0A0MT70	T cell receptor beta joining 2-6	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.71	IGHJ3	A0A0J9YW14	Immunoglobulin heavy joining 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72	TRBJ2-7	A0A0A0MT78	T cell receptor beta joining 2-7	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72	TRBJ2-5	A0A0A0MTA4	T cell receptor beta joining 2-5	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72	IGHJ4	A0A0J9YVS3	Immunoglobulin heavy joining 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.73	TRBJ1-3	A0A0J9YWP8	T cell receptor beta joining 1-3	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.74	TRBJ2-4	A0A0A0MT87	T cell receptor beta joining 2-4	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.74	TRBJ2-2	A0A0A0MT94	T cell receptor beta joining 2-2	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.74	TRBJ1-4	A0A0J9YXG5	T cell receptor beta joining 1-4	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.75	TRDJ4	A0A075B6Z4	T cell receptor delta joining 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.76	TRBJ2-3	A0A0B4J200	T cell receptor beta joining 2-3	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.76	TRDJ1	A0A075B706	T cell receptor delta joining 1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.77	TRBJ1-5	A0A0J9YXM7	T cell receptor beta joining 1-5	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.77	LOC128092247	A0A6Q8PFQ6	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.79	TRGJ1	A0A075B6S0	T cell receptor gamma joining 1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.79	TRGJ2	A0A5H1ZRR4	T cell receptor gamma joining 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.79	IGHJ5	A0A0J9YVP9	Immunoglobulin heavy joining 5 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.83	TRBJ2-1	A0A0A0MTA7	T cell receptor beta joining 2-1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.86	TRDJ2	A0A075B6V6	T cell receptor delta joining 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.86	TRBJ1-6	A0A0J9YWX3	T cell receptor beta joining 1-6	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.87		A0A8I5QKY2	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.87	TRAJ14	A0N4Z3	T cell receptor alpha joining 14 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.88	MIR155HG	C0HMA1	HSPA8-interacting micropeptide miPEP155	0.0-2.0 kDa	Actin family	configured audit family match: Actin family
1.89	TRAJ20	A0A075B6Z1	T cell receptor alpha joining 20 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.89	TRAJ43	A0A075B6V1	T cell receptor alpha joining 43 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.91	TRAJ27	A0A075B6W6	T cell receptor alpha joining 27 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.91	IGHJ1	A0A0C4DH62	Immunoglobulin heavy joining 1	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.91	TRAJ35	A0A075B6Y2	T cell receptor alpha joining 35 (non-functional) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.92	TRAJ49	A0A075B6Y0	T cell receptor alpha joining 49 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.93	TRAJ29	A0A075B711	T cell receptor alpha joining 29 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.95	TRAJ47	A0A075B6Y5	T cell receptor alpha joining 47 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.96	TRAJ59	A0A075B6V9	T cell receptor alpha joining 59 (non-functional) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.97	TRAJ11	A0A075B6Y8	T cell receptor alpha joining 11 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.97	TRAJ31	A0A075B700	T cell receptor alpha joining 31	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.01	TRAJ22	A0A075B6Z0	T cell receptor alpha joining 22 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.01	TRAJ36	A0A087WU04	T cell receptor alpha joining 36 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.01	TRAJ50	A0A075B6X7	T cell receptor alpha joining 50 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.02	TRAJ6	A0A075B6Z3	T cell receptor alpha joining 6 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.02	TRAJ33	A0A075B6W3	T cell receptor alpha joining 33 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.03	TRAJ17	A0A075B6W8	T cell receptor alpha joining 17 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.04	TRAJ41	A0A075B702	T cell receptor alpha joining 41 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.04	TRAJ54	A0A075B712	T cell receptor alpha joining 54 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.04	TRAJ9	A0N4Z8	T cell receptor alpha joining 9 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.05		A0AAQ5BH44	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.05	IGHJ2	A0A0J9YWN2	Immunoglobulin heavy joining 2 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.05	TRAJ42	A0A075B6Y9	T cell receptor alpha joining 42	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.06	TRAJ5	A0A075B6V8	T cell receptor alpha joining 5 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.06	TRAJ7	A0A075B714	T cell receptor alpha joining 7 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.06	TRAJ57	A0A075B704	T cell receptor alpha joining 57 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.07	TRAJ44	A0A075B6X8	T cell receptor alpha joining 44 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.08	TRAJ26	A0A075B6V7	T cell receptor alpha joining 26 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.08	TRAJ25	A0A075B6Z8	T cell receptor alpha joining 25 (non-functional) (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.09	TRAJ45	A0A075B6X0	T cell receptor alpha joining 45 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.09	TRAJ37	A0A087X096	T cell receptor alpha joining 37 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
2.09	TRAJ39	A0A075B710	T cell receptor alpha joining 39 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.09	TRAJ30	A0A075B6Y1	T cell receptor alpha joining 30 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.10	TRAJ13	A0A075B709	T cell receptor alpha joining 13 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.10	TRAJ38	A0A075B6W7	T cell receptor alpha joining 38 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.10	TRGJP1	A0A0A0MT97	T cell receptor gamma joining P1 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.10	TRAJ3	A0A075B6Y3	T cell receptor alpha joining 3	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.11	TRAJ4	A0A075B6Z5	T cell receptor alpha joining 4 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.11	TRGJP2	A0A0A0MT84	T cell receptor gamma joining P2 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.11	TRAJ56	A0A075B6Z2	T cell receptor alpha joining 56 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.12	TRAJ46	A0A075B6W4	T cell receptor alpha joining 46 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.12	TRAJ21	A0A075B6V2	T cell receptor alpha joining 21 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.13	TRAJ24	A0A075B6Z9	T cell receptor alpha joining 24 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.13	TRGJP	A0A0C4DH63	T cell receptor gamma joining P (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.14	TRAJ32	A0A075B6X3	T cell receptor alpha joining 32 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.14	TRAJ16	A0A075B6V0	T cell receptor alpha joining 16 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.14	TRAJ12	A0A075B6U8	T cell receptor alpha joining 12 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.15	TRAJ10	A0N4Z7	T cell receptor alpha joining 10 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.15	TRAJ23	A0A075B6U7	T cell receptor alpha joining 23 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.16	TRAJ34	A0N4X5	T cell receptor alpha joining 34 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.16		A0A8I5KYW3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.17	TRAJ48	A0A075B6V3	T cell receptor alpha joining 48 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.17	TRAJ53	A0A075B6W9	T cell receptor alpha joining 53 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.17	TRAJ28	A0A075B6X4	T cell receptor alpha joining 28 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.17	MT-RNR1	A0A0C5B5G6	Mitochondrial-derived peptide MOTS-c	2.0-4.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
2.18	TRAJ40	A0N4X2	T cell receptor alpha joining 40 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.18	PPY2P	Q9NRI7	Putative pancreatic polypeptide 2	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.19	TRAJ1	A0A075B713	T cell receptor alpha joining 1 (non-functional) (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.21	TRAJ2	A0A075B6U9	T cell receptor alpha joining 2 (non-functional) (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.22	TRAJ52	A0A075B6W2	T cell receptor alpha joining 52 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.24	TRAJ61	A0A075B708	T cell receptor alpha joining 61 (non-functional) (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.26	TRDJ3	A0A075B6W1	T cell receptor delta joining 3 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.29	TRAJ19	A0A075B6Y4	T cell receptor alpha joining 19 (non-functional) (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.37	IGLJ6	A0A0A0MT93	Immunoglobulin lambda joining 6 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.38	TRAJ18	A0A075B6X9	T cell receptor alpha joining 18 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.38		A0A8V8TPC4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.39	LOC139427322	A0AAQ5BH39	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.43		A0A8V8TNB9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.44	LOC128092251	A0A6Q8PHA8	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.45	IGHJ6	A0A0J9YVP2	Immunoglobulin heavy joining 6 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.57		A0AAQ5BIC9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.65	MTRNR2L7	P0CJ74	Humanin-like 7	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.66	MTRNR2L12	P0DMP1	Humanin-like 12	2.0-4.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38δ western blot	40.0-42.0 kDa	ORI reported p38δ bands were compiled from a source image representing β-actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38δ-labeled bands were compiled from a β-actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_GLUT4	Nrf2, NF-kB p65, PPARα/δ, Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	UP000005640_9606.fasta
Marker file	human_marker_genes.tsv
Signaling-prefix file	human_signaling_prefixes.tsv
Family-pattern file	human_family_patterns.tsv
Species/report label	Homo sapiens